

Selecting The RIGHT MICROCONTROLLER For EMBEDDED APPLICATIONS



Ronak Desai
is system
engineering
manager
at Cypress
Semiconductor,
Bengaluru

This article discusses the criteria for selecting the right microcontroller (MCU) for different embedded applications. It also discusses the design challenges and system limitations of MCUs in embedded applications.

Application in home appliance industry

The home appliance industry uses 8/16/32-bit MCU based circuitry for motor control and TRIAC/LED/LCD drive applications. The MCU controls and manages all functions and features of the home appliances. When you press Start button, inputs go to the MCU from the front-panel keypad and the MCU starts the three-phase brushless DC (BLDC) motor or the permanent magnet

synchronous (PMSM) motor. Motor speed is varied and controlled as per your input from the front-panel keypad.

The MCU uses either internal or external serial EEPROM (I2C/SPI based) to store old data. It uses a real-time clock for displaying accurate time. Temperature measurement is done using an onboard resistance temperature detector, thermistor or thermocouple based temperature-sensing device.

The MCU uses an external analogue-to-digital converter (ADC) and amplifiers for different analogue inputs from sensors, temperature sensors and battery. It uses external signal conditioning, comparators and gate-driver circuitry for driving and controlling the three-phase BLDC/PMSM motor. The MCU receives remote control inputs through an infrared receiver (at 38kHz input).

The MCU uses external buffer-driver circuitry to drive 7-segment LED/LCD/graphical display. Typically, a 7-segment LED/LCD/graphical display with backlight is used for showing the temperature, battery voltage, speed and error/warning messages. The MCU also interfaces with onboard peripherals like I2C/SPI and external peripherals like UART/USB for communication.

Application in automotive industry

The current automotive industry uses 16- to 32-bit MCU based circuitry for e-bikes. The circuitry controls and manages all functions and features of the automobiles. Once you use the ignition key to start the automobile, inputs go to the MCU. This starts the three-phase brushless automotive mo-

Fig. 1: Microcontrollers in home appliance applications

